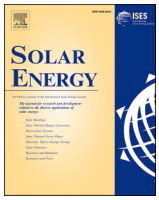
Solar Energy 274 (2024) 112588



Contents lists available at ScienceDirect

Solar Energy

journal homepage: www.elsevier.com/locate/solener





A DC arc fault location method for PV systems based on redundant antenna array and ellipse algorithm

Liangshi Lin [a], Wei Gao [a,b,*], Gengjie Yang [a]

[a] College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China
[b] Fujian Province University Engineering Research Center of Smart Distribution Grid Equipment, Fuzhou, China

ARTICLE INFO

Keywords:
Photovoltaic systems
Arc fault location
Redundant antenna array
Ellipse algorithm
DBSCAN

ABSTRACT

DC arc faults are major causes of electrical fires in photovoltaic (PV) systems. During the operation and maintenance of these systems, it is essential not only to identify arc faults but also to determine their exact locations accurately. To address the issue of DC arc fault localization in PV systems, this study investigates the electromagnetic radiation (EMR) characteristics of fault arcs and proposes a method for DC arc fault localization using the redundant antenna array and the ellipse algorithm. Firstly, during arc combustion, the EMR signals collected by antennas are subjected to median filtering to calculate the root mean square (RMS), which serves as the signal strength. An artificial neural network (ANN) model is constructed, which uses the signal strength and irradiance to predict the distance between the fault point and the receiving point. Subsequently, various redundant antenna array configurations are evaluated to assess the impact of different antenna quantities and layouts on localization accuracy. Once the optimal layout is determined, the three antennas with the strongest signal are selected. Their coordinates, along with the predicted distances to the fault point, are input into the ellipse algorithm, which is improved by trilateration, to obtain the locations of arc faults. Finally, the density-based spatial clustering of applications with noise (DBSCAN) method is used to fuse multiple measurement results, eliminate interference, and confirm the final fault coordinates. Experimental results demonstrate that the proposed location method exhibits excellent positioning capability and adaptability, with an average positioning error of 0.365 m.

1. Introduction

Photovoltaic (PV) power generation realizes the conversion of solar energy into electrical energy, reducing the reliance on conventional energy sources for electricity supply. It has garnered widespread attention from countries around the world [1–5]. PV power generation system comprises components such as PV arrays, combiner boxes, and PV inverters [6]. There are numerous interconnections and joints on the DC side of the arrays, which can easily suffer from insulation damage and poor contacts in outdoor environments, leading to the occurrence of DC arc faults, with series arc faults being particularly prominent [7,8]. However, conventional protection methods are unbale to detect the location of arc faults. Therefore, tracking the location of arc combustion and conducting maintenance work have become pressing research topics.

Most existing research primarily focuses on the detection and identification of DC arc faults. This is accomplished by extracting time-domain, frequency-domain, or time–frequency-domain features from the voltage and current during arc fault events and applying methods such as artificial intelligence and threshold-based techniques for feature recognition and classification. For example, Lu et al. [9] proposed a cross-domain DC series arc fault detection framework based on light-weight transfer convolutional neural network with adversarial data augmentation (LTCNN-ADA), more comprehensive transfer scenarios were explored and studied, including different types of power sources and inverters. Qian et al. [10] analyzed the current noise spectrum before and after the occurrence of DC arc faults, it was found that the adjacent multi-segment spectral similarity (AMSSS) characteristic could identify the event of the mark. Wang et al. [11] proposed fault detection methods based on voltage characteristic energy amplitude and phase mapping distribution distance to meet the fault detection for different load types.

During the formation of a fault arc, it exhibits high-frequency conduction currents and electromagnetic radiation (EMR), along with

* Corresponding author at: College of Electrical Engineering and Automation, Fuzhou University, Fuzhou, China; Fujian Province University Engineering Research Center of Smart Distribution Grid Equipment, Fuzhou, China.
E-mail address: gaowei0203@fzu.edu.cn (W. Gao).

https://doi.org/10.1016/j.solener.2024.112588
Received 27 October 2023; Received in revised form 29 March 2024; Accepted 29 April 2024
Available online 13 May 2024
0038-092X/© 2024 International Solar Energy Society. Published by Elsevier Ltd. All rights reserved.

macroscopic physical characteristics such as arc sound and arc light. Collecting, statistically analyzing, and calculating these features enable a more precise localization of the arc's occurrence [12]. For example, using externally installed capacitive current information for determination. Xiong et al. [13] proposed a novel method with the polarity and amplitude of capacitor currents. The integration of the frequency spectrum of capacitor currents is analyzed and identified as crucial attribute to differentiate DC series arc faults from other system changes. But it does not fully consider the effects of inverter start-up and environmental contributions to it. Akbar et al. [14] paralleled capacitors on PV arrays and extracted capacitor current signals to analyze series DC arc faults. Subsequently, by combining fast Fourier transform and artificial neural network signal processing techniques, the fault location is obtained. It is possible to acquire precise time–frequency-domain information from parallel capacitors, but this necessitates the installation of a relatively large number of capacitors across various substrings or branches. Additionally, the exploration of fault localization for DC bus systems is also discussed. Wu et al. [15] conducted theoretical analysis on extended spectra and time-domain reflection, proposing an extended spectrum time-domain reflection method based on the equivalent distributed parameters model of transmission lines. It fills the gap in detection and localization of DC bus arc faults in PV systems. In order to locate the fault of the solar power unit in the wind-solar energy storage system, a variational mode decomposition (VMD) algorithm in [16] is adopted to decompose the voltage signal of the same DC load under different working conditions, and the decomposition result is converted into a threshold value. It compared the time–frequency-domain features and threshold characteristics under different operating conditions for fault detection and localization. This method can adaptively determine the relevant frequency bands, but VMD has multiple control parameters that need to be carefully configured.

Compared to other physical characteristics, the higher-frequency EMR signals of arcs exhibit stronger resistance to interference, and with the use of suitable measurement antennas, sufficient sensitivity can be achieved. Some researchers have utilized EMR signals for arc faults localization, the existing methods mainly use neural networks to replace traditional ranging models for distance prediction. Li et al. [17] adopted neural networks and data augmentation techniques to estimate arc distance, presenting a planar localization method using only two detection points, which exhibited inherent immunity to normal current fluctuations (NCFs). The triangular localization method is easy to implement but is limited when facing more complex scenarios. For instance, it can only cover one direction and relatively short distances. Zhao et al. [18] investigated a Bayesian regularized neural network (BRNN) algorithm to establish an EMR model, estimating the distance between the arc and antenna through BRNN, and using antenna topology and the estimated distance to calculate the spatial coordinates of the arc. It involves comparing pretest data with data from unknown locations, so that this method has the issue of requiring frequent database updates.

Furthermore, it is worth mentioning that utilizing of EMR signals for fault source localization has been extensively studied in the field of high-voltage equipment's partial discharge (PD). Some of achievements in this area are also applicable to fault localization of PV arcs. However, PD typically occurs in switched equipment with higher discharge voltages and higher EMR signal frequencies. Compared to EMR signals of PV arcs, PD signals are easier to be detected due to their higher amplitude and frequency. Jangjoo et al. [19] used ultra-high-frequency (UHF) based on time-of-arrival (TOA) differential equations to locate PD in power transformers. They determined the optimal sensor placement on the transformer by measuring the distance between hyperbolic surface vertices and intersections, thereby reducing the impact of non-line-of-sight (NLOS) propagation, reflections, and refractions on electromagnetic waves. Chaisang et al. [20] introduced commonly used fingerprint-based positioning algorithms from indoor localization into PD localization. They analyzed measurement data using received signal strength, fingerprint recognition methods, and matched filters. These methods for locating the PD can be used as references, but cannot be fully applied.

In existing positioning methods, EMR signals are affected by fading and require high-speed acquisition instruments to collect and restore signals. The sampling conditions are rather stringent. The traditional trilateral positioning method often faces the problem of being unable to accurately calculate the positioning point. In addition, the environmental factors that affect the EMR signal of PV arrays are also complex. In response to the above issues, this paper proposes a new DC arc fault location method for PV arrays, with the following contributions:

1) Replace the traditional location method, trilateration, with the ellipse algorithm to solve the problem of multiple solutions and no solution, and obtain approximate solutions with smaller errors.
2) Construct the concept of redundant antenna array, analyze the influence of the number and layout of antennas in the array on calculation results and convenience, and obtain the optimal antenna layout scheme to reduce the impact of fading.
3) Use density-based spatial clustering of applications with noise (DBSCAN) to fuse multiple measurement results, eliminate the interference of noise signals and unstable arc data, and further improve positioning accuracy.

The remainder of this paper is organized as follows. Section 2 introduces the experimental platform and analyzes the EMR signal characteristics in PV systems. Section 3 presents ellipse location algorithm and redundant antenna array. The proposed algorithm is verified with measured data and compared with other methods in Section 4. Eventually, Section 5 offers the conclusion of this paper.

2. Analysis of PV DC arc EMR signal

2.1. Experimental platform

A typical PV system comprises several key components, including PV array, combiner box, inverter, and protection device, as shown in Fig. 1. It shows the potential locations of arc faults, denoted as ① to ⑤, corresponding to the DC busbars ends, the head, the middle and the end of subarrays, and between the PV subarrays, respectively. Considering that the wiring within the subarrays is often concealed beneath the array and relatively well hidden, thus possessing a higher degree of concealment, while the busbar wiring layout depends on the inverter's installation position and may not be as concealed, this paper mainly focuses on the localization of arc faults within (or between) the PV array subarrays, namely the four positions: ② to ⑤.

The PV array structure used in the experiment is configured as 2×12, meaning 12 modules are connected in series, and then 2 strings are connected in parallel (outlined by red and yellow dashed lines). The on-site scene is depicted in Fig. 2, with a capacity of 6.48 kWp. Some technical parameters are listed in Table 1, and the first three coefficients represent the values of the inverter in MPPT (maximum power point tracking) state. The structure of Fig. 2 is consistent with that of Fig. 1.

Arc faults are simulated through arc fault generator (AFG). Meanwhile, the specific structure and operation method of AFG can be found in [6].

The acquisition process of EMR signal in this article is illustrated in Fig. 3. The parameters of the antenna, amplifier and oscilloscope are listed in Table 2. In all experiments, the antennas and AFG are situated in the same plane.

2.2. Time- frequency- domain analysis of DC arc

The EMR signals are collected both during the normal and the arc fault condition. The sampling frequency is 1.0 GHz. The fast Fourier transform (FFT) is applied to observe the frequency-domain components of the signals, as shown in Fig. 4.

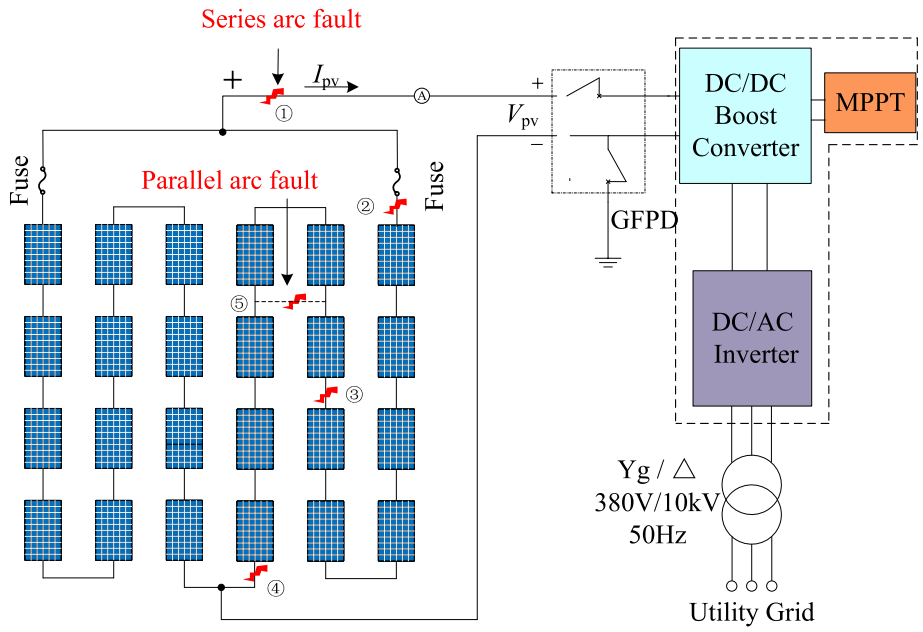


Fig. 1. Composition of PV system.

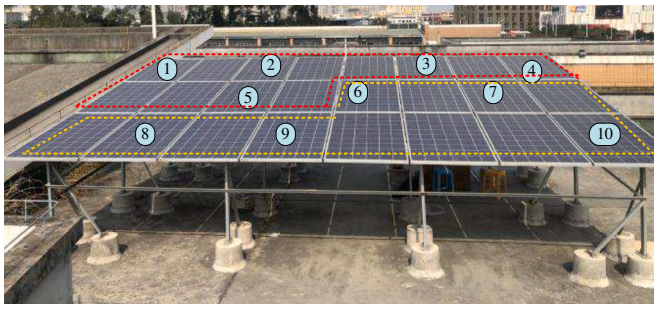


Fig. 2. Field scene graph map of PV system.

Table 1
Parameters of PV module under standard test conditions.

P_{mppt}	V_{mppt}	I_{mppt}	V_{oc}	I_{sc}
270 W	31.3 V	8.63 A	38.5 V	9.09 A

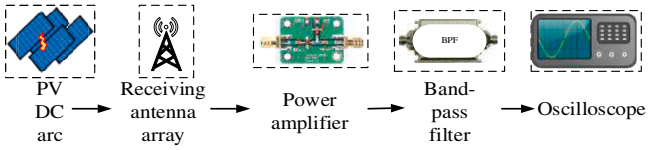


Fig. 3. Acquisition flowchart of EMR signal.

It can be observed that the frequency-domain of the arc predominately concentrates between 200 ~ 700 MHz. Therefore, the filtering frequency band of the bandpass filter selected in this article is set within this range, and the time-domain signal within this frequency band is applied as the data source for subsequent positioning.

Accurate modeling is of great significance for studying the characteristics of fault arcs [21]. Additionally, Jin et al. [22] further derived the relationship equation between power and arc current and arc length, suggesting that both arc length and current would affect the EMR intensity of the arc, with current having a great impact in comparison.

Table 2
Parameters of electromagnetic experimental equipment.

Equipment	Parament	Value
Antenna	Frequency band	20 ~ 500 MHz
	Gain	−5 ~ 2 dB
	Noise coefficient	≤7 dB
	Output impedance	50 Ω
Amplifier	Frequency band	50 MHz ~ 6 GHz
	Gain	20 dB
Filter	Frequency band	200 ~ 500 MHz
Oscilloscope	Bandwidth	2 GHz

During arc combustion period, the arc length exhibits randomness, making it difficult to accurately measure the actual length of the arc. Therefore, the impact of arc length is not considered in this paper, and instead approximated by using the electrode gap length of AFG, with the gap length being randomly assigned.

2.3. Determination and processing of EMR signals

In the field of indoor positioning, scholars often use the average value of signals received by the receiver within a certain time period to calculate signals. For example, Cappelli et al. [23] proposed using the average irradiance of indoor visible light as RSSI (received strength signal indicator) to reduce the impact of irradiance fluctuations on positioning results. For outdoor PV arrays, the current generated by PV modules varies with irradiance and is accompanied by intermittent pulse spikes. At this point, directly calculating the average value will introduce significant errors, so alternative mathematical statistical methods should be sought. To address this issue, the root mean square (RMS) value was studied instead [22]. The specific process involves receiving steady-state EMR signals with antennas, setting a sampling time window of five milliseconds, and calculating the RMS value within this five-millisecond waveform as a positioning feature indicator. But it does not consider the impact of pulse peaks on the subsequent positioning process. On the other hand, Yu et al. [24] proposed the collection of UHF EMR signals (300 ~ 1500 MHz) at a low sampling rate (2.70 MHz), eliminating the need for signal restoration and improving the algorithm's cost-effectiveness.

Furthermore, to investigate the variation of arc EMR signals, AFG

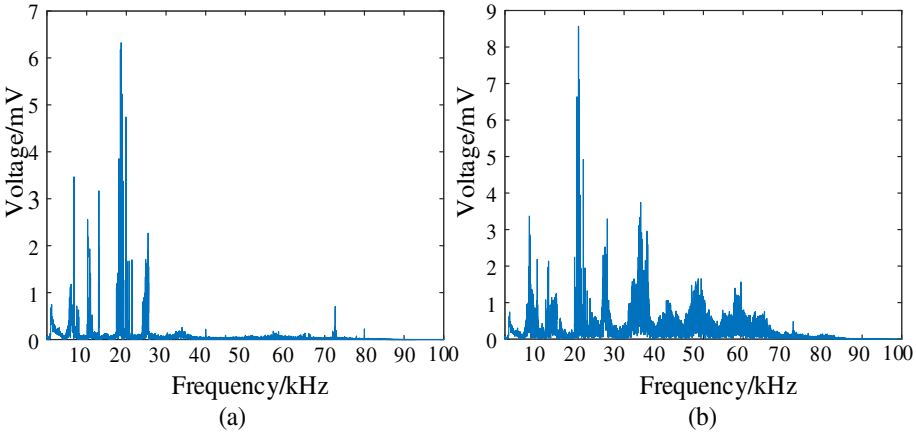


Fig. 4. EMR signal spectrum before and after fault occurrence. (a) background noise (b) series arc fault.

was employed to simulate the process of arc generation, as shown in Fig. 5, with a sampling rate of 200 kHz. Fig. 5 compares the current signal waveform and EMR signal waveform across four states of a DC arc: normal operation, initiating, stable combustion, and arc extinction. At each state, the current signal waveform exhibits distinct characteristics. In contrast, the arc EMR signal waveform shows a relatively consistent trend: as the arc ignites, the amplitude of the EMR signal rapidly increases, and upon the arc's extinction, the EMR signal promptly returns to its normal state, displaying minimal transition.

After an arc fault occurs in PV array, it is advisable to promptly disconnect the fault rather than immediately locating the source of the fault. The primary objective of fault location is to efficiently identify the fault point during subsequent operation and maintenance. In this paper, the waveform during the stable combustion state after the fault occurrence is chosen as the positioning signal.

The method for obtaining data involves the following steps: Firstly, EMR signals are collected during the stable combustion state of the arc with a sampling rate of 200 kHz. Subsequently, a random selection of waveforms is extracted, with each waveform consisting of a special number (a) and having a duration of five milliseconds. For each waveform, the median filtering method is introduced to eliminate pulse interference points. Finally, the RMS value is computed and served as E_{rms}.

$$E_{\text{rms}} = \sqrt{\frac{\sum_{v=1}^{N} X_v^2}{N}} \tag{1}$$

where X_v is the processed arc EMR signal with a length of N.

2.4. Prediction distance of fault location

In the field of wireless communication, fading refers to the phenomenon of signal attenuation caused by multipath transmission, which can be divided into large-scale fading (including path loss and shadow fading) and small-scale fading (including multipath and Doppler effects). Fading is a crucial factor in the propagation process of electromagnetic wave and plays a significant role in analyzing the quality and reliability of communication signal. Compared to the commonly used log-normal shadow model (LNSM), machine learning models offer better

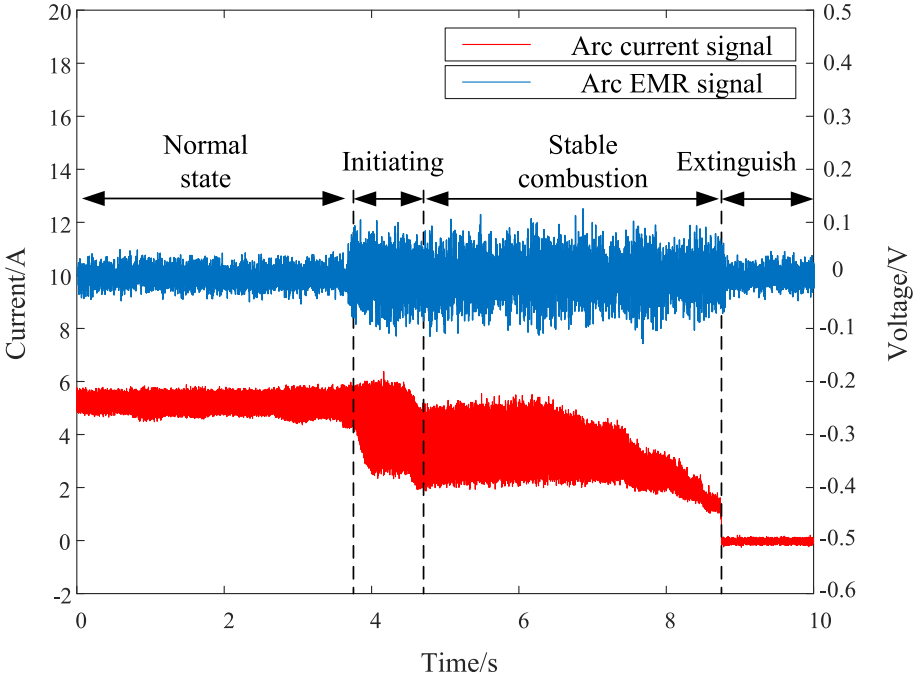


Fig. 5. Waveforms of arc current and EMR signal.

fitting performance and they are more powerful tools for analyzing fading. For example, different types of environmental features are extracted to construct an ideal path loss model, which can effectively cope with the development trends of wireless channel diversity, time-varying nature, and massive characteristics [25,26].

Since the arc current will affect the intensity of EMR signal, and the irradiance is the main factor to change the output current of PV system. When other factors remain unchanged, the greater the irradiance, the larger the output current, resulting in a smaller EMR signal. At the same time, distance is also a key factor affecting the strength of EMR signal. Specifically, the arc EMR density and irradiance were took as input to a BP neural network [22], predicting the distances from the observation points to fault location. Following the similar approach as described in these references, this paper incorporates E_{rms} and irradiance as input to ANN to predict distance values. The internal structures and parameters of the network have been adjusted to achieve better prediction results. The accuracy of distance prediction plays a crucial role in the positioning algorithm discussed in the next chapter, ultimately impacting the final positioning results.

3. Location method and strategy

3.1. Ellipse algorithm

3.1.1. Algorithm introduction

The traditional trilateration method is based on distance measurement algorithm to obtain the distances between an unknown node and three reference nodes. Then, using distance equations, the coordinates of the locating node will be solved. In this paper, the locations of three reference nodes, represented by the coordinates of three antennas, are denoted as $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$. The unknown node' location, which is the coordinate of the arc fault point, is denoted as $H(x_o, y_o)$. In the event of an arc fault, each of the three antennas receives EMR signals emitted by the arc combustion. Using the results fitted by ANN can obtain the distances from the arc fault point to antennas A, B and C as d_1, d_2 and d_3, respectively. With the known positions of three antennas and the distances from the arc fault point to the three antennas, it is possible to calculate the coordinate of the arc fault location.

In the ideal trilateration method, when d_1, d_2 and d_3 are accurately determined without errors, it is indeed possible to calculate the unique and correct coordinate $H(x_o, y_o)$ [27]. Using points A, B and C as the centers and d_1, d_2 and d_3 as the radii, circles are drawn as shown in Fig. 6 (a). At the same time, the intersection point of these circles represents H.

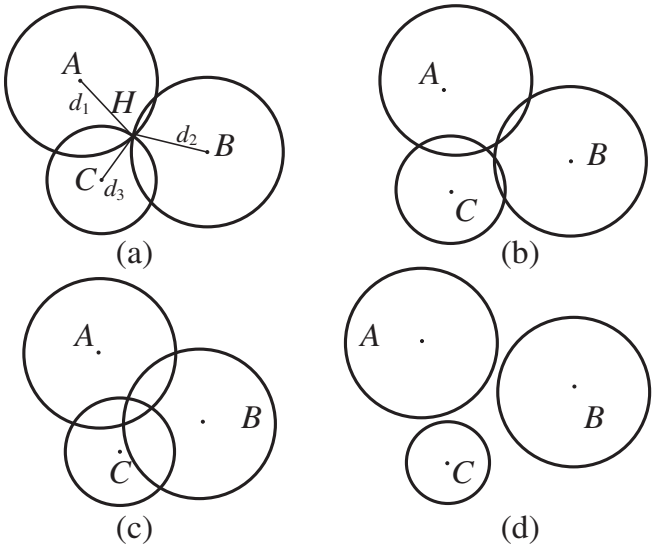


Fig. 6. Cases analysis of trilateral positioning method. (a) Intersection of three circles at one point (b) Intersection of three circles I (c) Intersection of three circles II (d) Three circles without intersection.

Complex environments can indeed affect the transmission effect of electromagnetic wave and introduce measurement errors. If the measured values are biased toward larger values, situations similar to those shown in Fig. 6 (b) and (c) may arise, where circles intersect between circles, resulting in a system of equations with multiple solutions. Conversely, If the measured values are biased toward smaller values, meaning that circles do not intersect between circles, as shown in Fig. 6 (d), the system of equations may have no solution.

Given the above circumstances, this study proposes an ellipse algorithm to solve the problems of multiple solutions and no solution in traditional trilateration method. Based on the introduction to the principles of the ellipse algorithm, two lemmas are illustrated [28,29].

Lemma 1. For the union of ellipsoids, it is possible to compute its Löwner-John ellipsoid, which is the smallest volume ellipsoid containing the set. Once the minimum volume ellipsoid containing the union of ellipsoids $\cup_{t=1}^{m}\varepsilon(q_t, Q_t)$ is determined, there exist variables J, b and λ_i that satisfy the convex optimization problem of semi definite programming (SDP):

$$\min \lg \det J^{-1}$$

subject to:

$$\lambda_t > 0$$

$$\begin{bmatrix} J^2 - \lambda_t Q_t^{-1} & Jb + \lambda_t Q_t^{-1} q_t & 0 \\ (Jb + \lambda_t Q_t^{-1} q_t)^{\mathrm{T}} & -1 - \lambda_t (q_t^{\mathrm{T}} Q_t^{-1} q_t - 1) & (Jb)^{\mathrm{T}} \\ 0 & Jb & -J^2 \end{bmatrix} \le 0, t = 1, \cdots, m. \quad (2)$$

after J and b are found,

$$q = -J^{-1}b, Q = (J^{\mathrm{T}}J)^{-1} \quad (3)$$

then, $\varepsilon(q, Q)$ can be determined.

Lemma 2. Find a maximum volume ellipsoid, which is located inside a non-empty bounded convex set. It is possible to determine the maximum volume ellipsoid within a given intersection of ellipsoids $\cap_{t=1}^{m}\varepsilon(q_t, Q_t)$. If there exist variables V, q and λ_i that satisfy the SDP convex optimization problem:

$$\min \lg \det V^{-1}$$

subject to:

$$\lambda_t > 0$$

$$\begin{bmatrix} 1 - \lambda_t & 0 & (q - q_t)^{\mathrm{T}} \\ 0 & \lambda_t I & V \\ q - q_t & V & Q_t \end{bmatrix} \ge 0, i = 1, \cdots, m. \quad (4)$$

after V and q are found,

$$Q = V^{\mathrm{T}}V \quad (5)$$

then, $\varepsilon(q,Q)$ can be determined. The two lemmas mentioned above satisfy affine invariance, and the conclusions in three-dimensional space also apply to two-dimensional planes, where the maximum cross-section of an ellipsoid projected onto a plane forms an ellipse [29]. Specifically, the steps of the ellipse algorithm proposed in this article are as follows:

1) Three antennas, labeled A, B and C, are set up along with the arc fault point H. After E_{rms} is input into ANN to fit the distances, the known information includes the coordinates of the three antennas and the distances from the arc fault point to each antenna.

2) With antennas A, B, and C as the centers and the respective distance from the antennas to the arc fault as the radii, create three circles using the following equation.

$$\varepsilon_n(x_n, Q_n) = \{x_o : (x_o - x_n)^T Q_n^{-1}(x_o - x_n) \le 1\} \tag{6}$$

where $Q_n = d_n^2 I$, $n = A, B, C$.

3) Lemma 1 is used to find the union between the two pairs of the three circles in order, and then find the smallest outsourcing ellipse to three ellipses, which are represented by the following formula.

$$\begin{aligned} \varepsilon_\alpha(x_\alpha, Q_\alpha) &\supseteq \varepsilon_A(x_A, Q_A) \cup \varepsilon_B(x_B, Q_B) \\ \varepsilon_\beta(x_\beta, Q_\beta) &\supseteq \varepsilon_B(x_B, Q_B) \cup \varepsilon_C(x_C, Q_C) \\ \varepsilon_\gamma(x_\gamma, Q_\gamma) &\supseteq \varepsilon_A(x_A, Q_A) \cup \varepsilon_C(x_C, Q_C) \end{aligned} \tag{7}$$

4) Lemma 2 is used to find the inner ellipse with the largest area of the intersection of the three ellipses in Step 3), expressed by the following formula.

$$\begin{aligned} \varepsilon_\lambda(x_\lambda, Q_\lambda) &\supseteq \varepsilon_\alpha(x_\alpha, Q_\alpha) \cup \varepsilon_\beta(x_\beta, Q_\beta) \\ \varepsilon_\mu(x_\mu, Q_\mu) &\supseteq \varepsilon_\gamma(x_\gamma, Q_\gamma) \cup \varepsilon_\lambda(x_\lambda, Q_\lambda) \end{aligned} \tag{8}$$

5) Finally, the coordinate of the center of the ellipse, $\varepsilon_\mu(x_\mu,Q_\mu)$, is set as the location of the arc fault.

3.1.2. Algorithm simulation

In this paper, the Ellipsoidal Toolbox of MATLAB is used for simulation, and the four cases shown in Fig. 6 are analyzed by drawing, as shown in Fig. 7. The blue line, the black line, and the purple line represents the circle, ellipse, and intersections made according to Step 2), Step 3), and Step 4) of Section 3.1.1, respectively. The orange line represents the ellipse made according to Step 4) of Section 3.1.1, and the asterisk represents the calculation result based on Step 5) of Section 3.1.1, which is presented by E. At last, the coordinates of the three circulars are A (-1, 4), B (4, 1.5), C (-0.7, -1), and the ideal intersection coordinate is H (0, 1).

The radius of the three circles and the localization errors for the four different scenarios are listed in Table 3. Reasonable radius lengths are set to produce the corresponding results for each scenario. The positioning errors are expressed by the absolute values of the distance differences (Euclidean distance) between E and H, where the positioning errors of case (a), (c), and (d) in Fig. 7 are less than 0.5, while the positioning errors of case (b) in Fig. 7 are slightly larger at 0.6483.

It is evident that the ellipse algorithm can provide an approximate solution when there are multiple solutions (with intersection) and no solution (without intersection) in the trilateration method, and the positioning errors are within an acceptable range, indicating the effectiveness of this algorithm.

3.2. Redundant positioning strategy

3.2.1. Limitation of localization using three antennas

There may be various obstacles under the PV array, including brackets, roofs, and various electrical equipment. This area can be considered as a NLOS environment, as shown in Fig. 8. Unfortunately, these obstacles will cause shadow fading and multipath effect of the EMR signals, affecting the ability of antennas to receive EMR signals.

Table 3
Parameters and errors of the ellipse algorithm.

	Fig. 7 (a)	Fig. 7 (b)	Fig. 7 (c)	Fig. 7 (d)
Circle A radii	3.1623	2.9	3.2	2
Circle B radii	4.0311	3.4	5.2	3
Circle C radii	2.1191	2.4	2.6	2
Positioning error	0.4901	**0.6483**	0.3859	0.4918

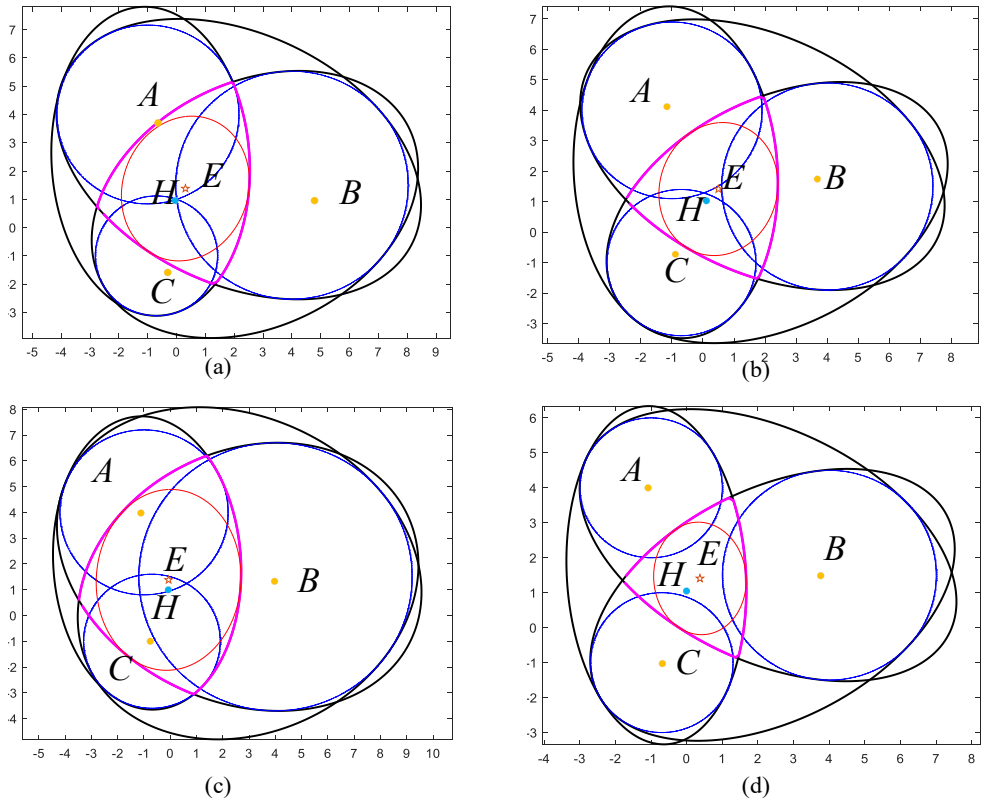


Fig. 7. Analysis of ellipse algorithm. (a) Intersection of three circles at one point (b) Intersection of three circles I (c) Intersection of three circles II (d) Three circles without intersection.



Fig. 8. Sketch map of obstacles under PV array.

For this reason, to analyze the impact of EMR signals, arc faults are set respectively in the PV array area and the open area, with a total of four locations, as shown in Fig. 9.

In the PV array layout area, PV array is projected vertically onto the ground, forming a rectangular shape with a length of 8 m and a width of 4.875 m. Using the longer side as the *x*-axis and the shorter side as the *y*-axis, with the lower left corner as the origin, a Cartesian coordinate system is established. Three antennas are placed with the following coordinates: $S_1(0, 4.875)$, $S_2(8, 2.4375)$, $S_3(0, 0)$. Moreover, at each of the four locations, AFG is operated to initiate arcing during the stable combustion phase. Data is collected within a two-second timeframe by sampling 10 waveforms, each lasting five milliseconds. These collected data are processed to obtain E_{rms} values. To reduce the influence of randomness or noise, the average E_{rms} value is calculated from these 10 E_{rms} values.

Subsequently, the antennas and AFG are moved to an open area. In this open area, a rectangular region of the same size is defined, maintaining the antenna array layout and coordinates consistent with the previous setup. Undoubtedly, the same method as described above is used to simulate arcing in the open area and obtain the average E_{rms} values. These two experiments are conducted under the conditions of an irradiance level between 489 to 512 W/m^2, a temperature between 25 and 27 ℃, and no other electromagnetic interference sources in the vicinity. The results are presented in Table 4.

It can be observed that EMR signal propagation is affected when compared to the open area in the vicinity of PV array. Antenna S_1 at location 1, antenna S_2 at location 4, and antenna S_3 at locations 1, 3, and 4 all experience significant signal attenuation. Specifically, under the conditions of irradiance ranging from 200 to 1200 W/m^2, and temperature between 15 and 55 ℃, with no external electromagnetic interference, the minimum measured E_{rms} value should not be less than 0.03 V. A value below this threshold can be considered as the antenna not receiving any signal. It can be seen that for antenna S_2 at location 4 under PV array, the measured average E_{rms} value is 0.0279 V, which is less than 0.03 V. Therefore, antenna S_2 detects little EMR signal at this location. In such case, the ellipse algorithm based on three antennas would fail to meet the requirement.

Table 4
Average E_{rms} (V) received from four locations of three antennas.

Antenna Location	S_1 PV	Open	S_2 PV	Open	S_3 PV	Open
1	**0.0598**	**0.0635**	0.1306	0.1308	**0.0611**	**0.0683**
2	0.0961	0.0962	0.0877	0.0891	0.0889	0.0894
3	0.0736	0.0740	0.0656	0.0663	**0.0596**	**0.0712**
4	0.1113	0.1126	**0.0279**	**0.0423**	**0.0991**	**0.1221**

3.2.2. Redundant antenna array

To address the issue mentioned above, it is necessary to increase the number of antennas, forming a redundant antenna array. In the PV array layout area, n antennas are arranged, each with corresponding coordinate. These antennas are numbered clockwise from the outermost to the innermost as $S_1, S_2, \ldots, S_n$, where $n \geq 4$. The top three antennas with the max E_{rms} values are selected:

$$E_{rmsi}, E_{rmsj}, E_{rmsk}, i, j, k \in n \tag{9}$$

make

$$E_{rmsi} \geq E_{rmsj} \geq E_{rmsk} \geq E_{rmsm} \tag{10}$$

where $m \neq i, j, k, m \in n$.

The positioning model is illustrated in Fig. 10, where the arc fault emits EMR signals to the surrounding area. Firstly, there are obstacles between antenna S_2 and S_7 and the arc. The signal is mainly affected by shadow fading, leading to a reduction in amplitude. Secondly, there may be multiple propagation paths between antenna S_4 and the arc due to the obstacles. The signal is mainly affected by multipath effects. Then antenna S_6 is far from the arc, so the signal is mainly affected by path loss.

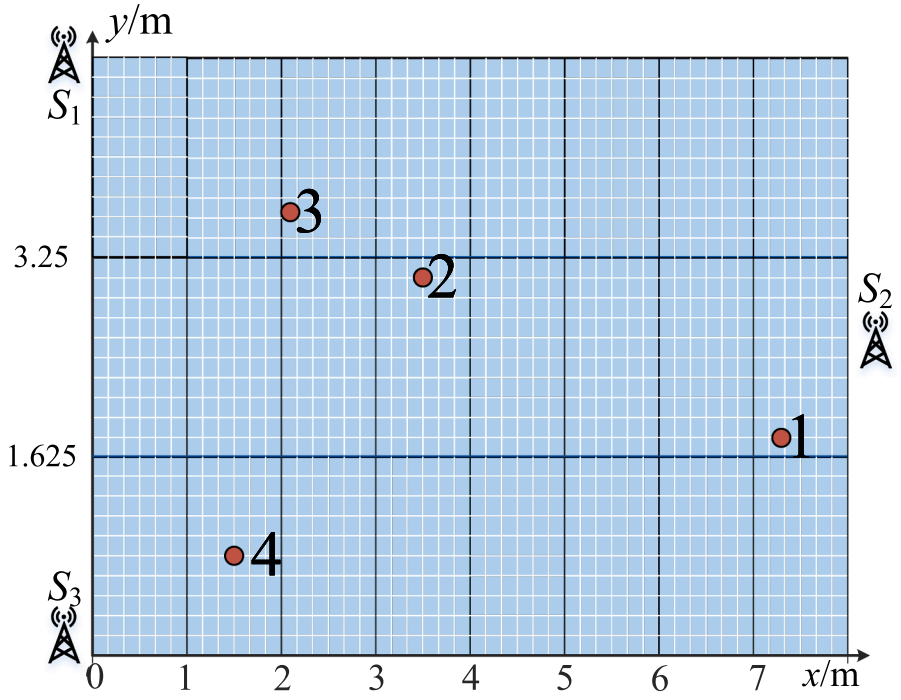


Fig. 9. Layout of three antennas.

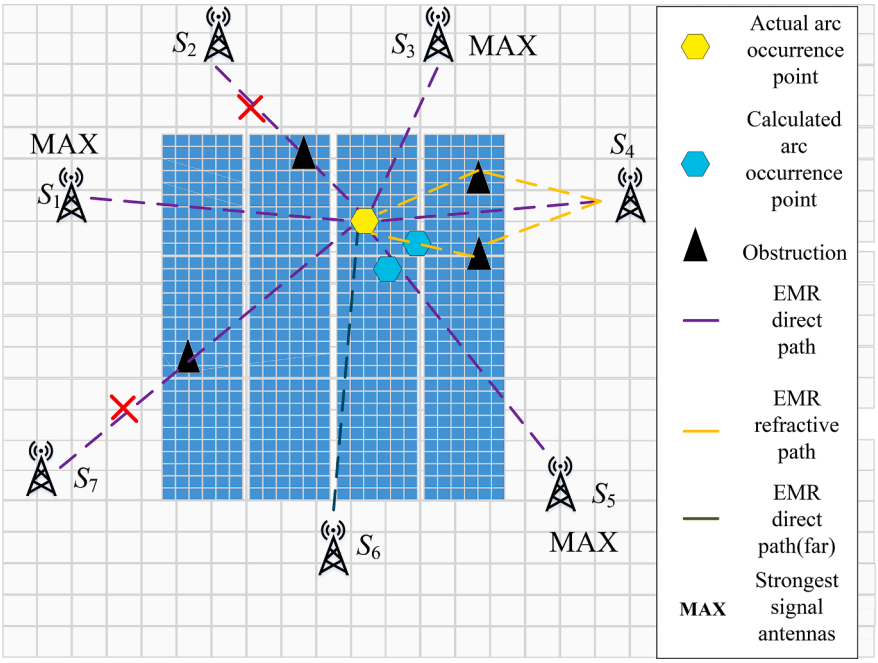


Fig. 10. Example of location model.

Finally, after comparison, S_1, S_3 and S_5 are the three antennas with the strongest signals, receiving a significant portion of the direct signal. The final positioning result corresponds to the arc occurrence point calculated by the algorithm, as indicated in the diagram. However, there may be a positional deviation from the actual arc occurrence point.

In addition, the performance of positioning can also be influenced by the arrangement of the antenna array [30]. Symmetrical layouts tend to have relatively better performance and are widely used in engineering applications. This study will discuss the arrangement of redundant antenna array. Several commonly used layout options are played in Fig. 11, which are named based on a pattern: the first number represents the number of antennas, and the letter R, D, and H respectively denotes rectangular, diamond, and hexagonal structure. For example, "4R" represents the layout with four antennas in the rectangular structure. The coordinate of each antenna is listed in Table 5. As the number of antennas increases, the computational complexity and time overhead grow exponentially, while the improvement in accuracy may not be significant. Generally, layout with six antennas is sufficient to meet

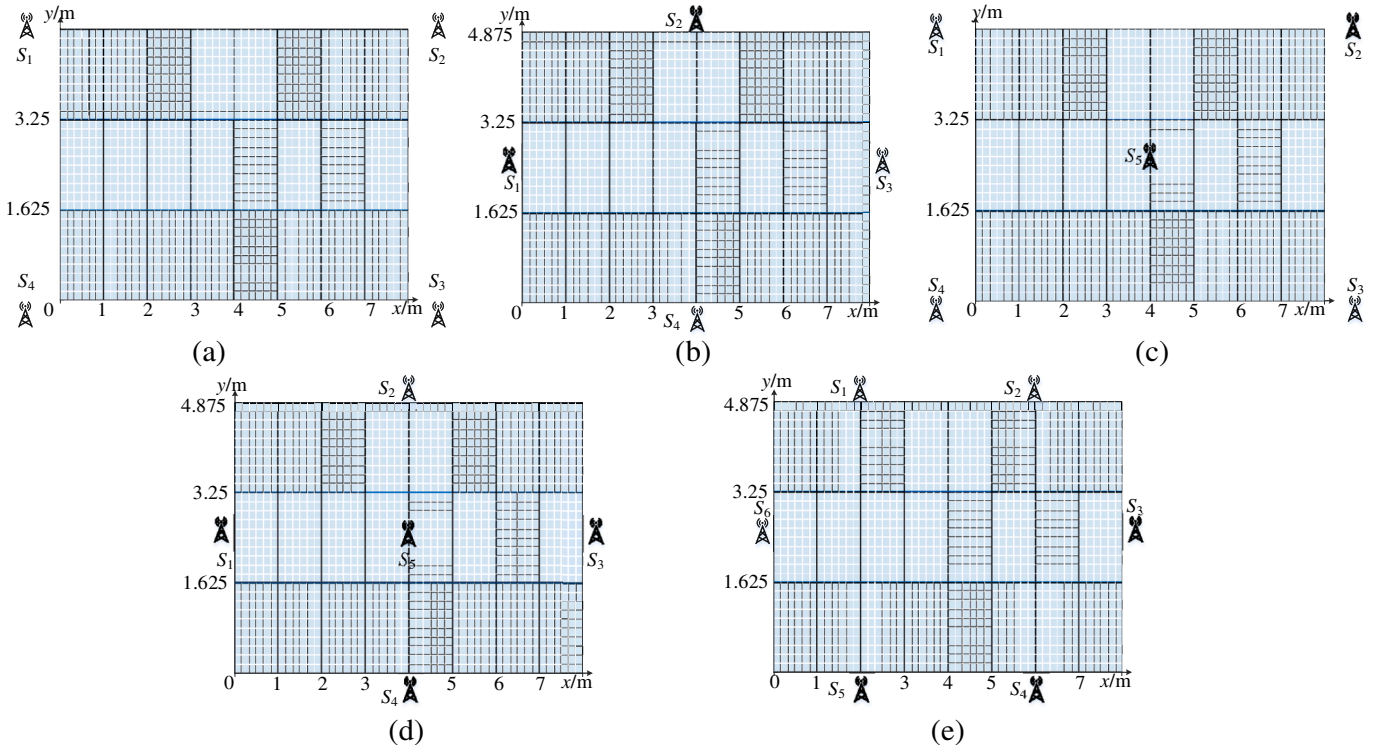


Fig. 11. Antenna layout. (a) 4R (b) 4D (c) 5R (d) 5D (e) 6H.

Table 5
Coordinates of each antenna for five layouts.

Anteanna	4R	4D	5R	5D	6H
S_1	(0,4.875)	(0,2.4375)	(0,4.875)	(0,2.4375)	(2,4.875)
S_2	(8,4.875)	(4,4.875)	(8,4.875)	(4,4.875)	(6,4.875)
S_3	(8,0)	(8,2.4375)	(8,0)	(8,2.4375)	(8,2.4375)
S_4	(0,0)	(4,0)	(0,0)	(4,0)	(6,0)
S_5	–	–	(4,2.4375)	(4,2.4375)	(2,0)
S_6	–	–	–	–	(0,2.4375)

practical application requirement.

3.3. Analysis of DBSCAN clustering

During a single arc development cycle, collecting multiple signal samples and effectively fusing the results can improve the accuracy of fault point localization. In this paper, DBSCAN clustering method is used to fuse and analyze the multiple measurement results. DBSCAN is a representative density-based clustering algorithm that defines clusters as the largest set of points connected by density. It can partition regions with sufficiently high density into clusters and can discover clusters of arbitrary shapes in spatial databases with noise. Compared to traditional K-means method, DBSCAN does not require prior knowledge of the number of clusters to be formed, and can identify clusters of arbitrary shapes and recognize noise points.

DBSCAN has two input parameters: the neighborhood radius *Eps* and the minimum number of data objects in the neighborhood *MinPts*. Its output is density-connected clusters. Within radius *R*, data points with a count greater than or equal to *MinPts* are considered core points. Points that are not core points but are within the neighborhood of a core point are considered border points, whereas points that are neither core points nor border points are classified as noise points. The processing flow of the clustering algorithm is as follows:

1) Randomly select a point *p* (neither assigned to a class nor specifically as a periphery point), and calculate whether the selected data object point is a core point for the parameters *Eps* and *MinPts*. If so, find all data object points reachable from the density of *p* to form a cluster.
2) If the selected data object point *p* is an edge point, select another data object point.
3) Repeat Steps 1) and 2) until all points are within the cluster (core or edge points) or are noise points.

Clusters with more sample points are selected, and clusters with less sample points and noise points are abandoned [31]. Referring to K-means algorithm, the mean value of each sample point is calculated and taken as the clustering center.

3.4. Positioning process

This paper proposes a novel DC arc fault location method for PV array. The specific steps are as follows, the flowchart is shown in Fig. 12.

1) Using a single antenna to collect arc EMR signals at different distances and irradiances at a sampling rate of 200 kHz, and obtain E_{rms} values after processing.
2) Use E_{rms} and irradiance to train ANN in order to predict the output distance of the model.
3) Set up a Cartesian coordinate system with PV array as the plane, determine the number and layout of redundant antenna array, and place the antennas.
4) Convert the arc EMR signal collected by the antenna array into E_{rms}, calculate the average of multiple time windows, and select the antenna according to the order of E_{rms} size.
5) Input the E_{rms} of the three antennas together with the corresponding irradiance into the trained ANN to obtain the predicted distance from the arc to the antenna array, and apply the ellipse algorithm to obtain the location of the arc fault point.
6) Conduct DBSCAN clustering analysis to further reduce errors and obtain the final coordinates of the arc fault point, completing the location.

4. Experimental testing

4.1. Determination of model parameters

4.1.1. Collection of dataset

The AFG is interposed between PV modules, and dataset samples are collected during the stable combustion of the arc. Moreover, the oscilloscope sampling frequency is set to 200 kHz, and the irradiance range is between 200 and 1100 W/m^2.

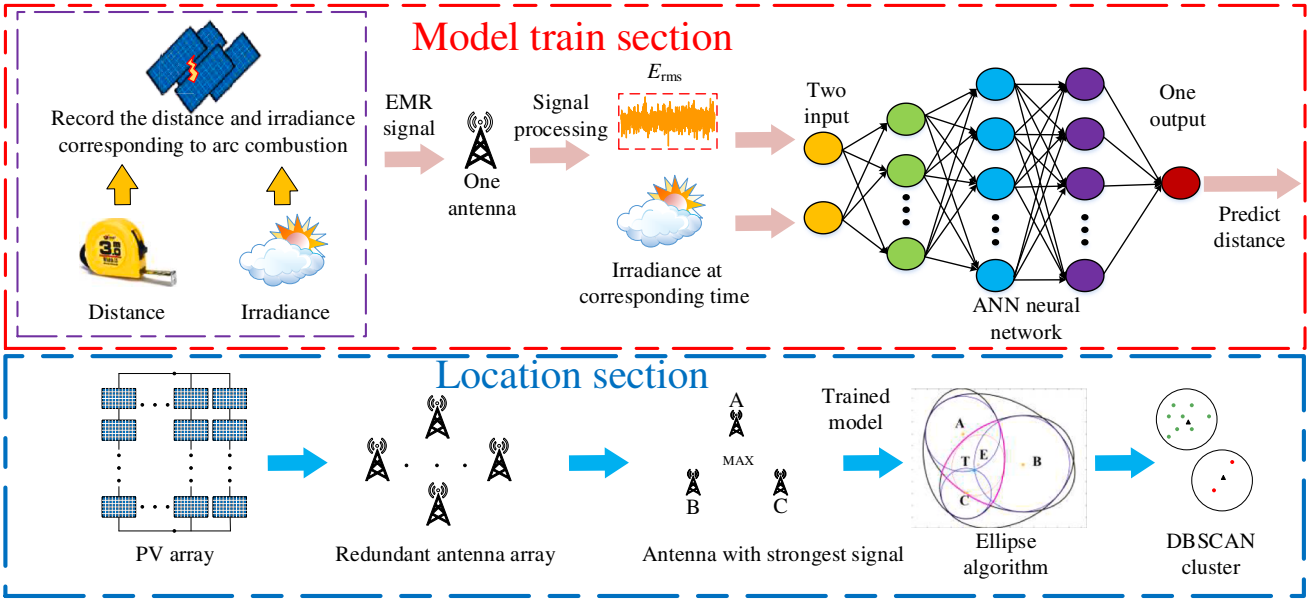


Fig. 12. Integral process of proposed method.

With a step size of 0.5 m, the distance between the antenna and AFG is gradually increased. Using one antenna, a large amount of arc's EMR signal data is collected at different irradiance levels and distances, thus forming the training dataset for the ANN. In this study, the maximum distance from the antenna to the arc fault source under the PV array is 8.942 m, and the farthest effective distance at low irradiance in the open area where the antenna could receive EMR signals is 11.389 m. Therefore, the range of training samples is set from 0 to 9 m. As for the test dataset, antennas are placed at the positions of the antenna layout shown in Fig. 11. With varying step sizes, EMR signals are collected at different irradiance levels and distances, with the distance range still being 0 to 9 m.

In a word, a total of 22,350 samples are obtained in this paper. Out of these, 16,008 samples are used to train the ANN network, with a ratio of 5:1 between the training set and validation sets. The remaining 6,342 samples are investigated to test the model's performance. The number of iterations is set to 3000, and the Adam is selected as the optimizer.

4.1.2. Predictive effects of ANN

Test samples are fed into the trained ANN to test the predictive power of the model. Mean absolute error (MAE) is used to characterize the prediction results, as shown in the Fig. 13. It can be seen that among the 19 predicted locations, MAE is smaller when the distance is less than 5 m. When the distance is greater than 7 m, the MAE is larger, and even when the distance is 9 m, the MAE is 0.512 m, greater than 0.5 m. For this result, it can be interpreted that the farther away the antenna is from the arc, the smaller the arc EMR signal received, and the more susceptible it is to interference from other electromagnetic signals around. The average error of the prediction results is 0.308 m, which shows that the prediction effect is good.

4.1.3. Time window

To establish the appropriate time window for capturing EMR signals, this paper uses a representative distance of 0.25 m between the antenna and the arc. Forty groups of samples are selected from the training set, with each group representing two distinct irradiance levels. The difference of RMS value of arc EMR signals collected with four time windows of 1 ms, 2.5 ms, 5 ms and 7.5 ms at the same distance and different irradiance are shown in Fig. 14. The red mark represents the RMS value distribution of arc EMR signals when the distance is 0.25 m and the irradiance is 359 W/m^2. The blue mark represents the RMS value distribution of arc EMR signals when the distance is 0.25 m and the irradiance is 510 W/m^2.

When the distance between the arc and the antenna is constant, the RMS values of the arc EMR signal under different irradiance gradually increase with the increase of the time window. Under the same irradiance, the dispersion of EMR samples with different arcs decreases with the increase of time window. In other words, the larger the time window, the more different the RMS values of arc EMR signals obtained under different irradiance. However, too large time window will not only increase the calculation burden, but also weaken the randomness of the arc. Overall, the time window size in this paper is set to 5 ms. It should be noted that in the 5 ms time window, the irradiance can be regarded as constant.

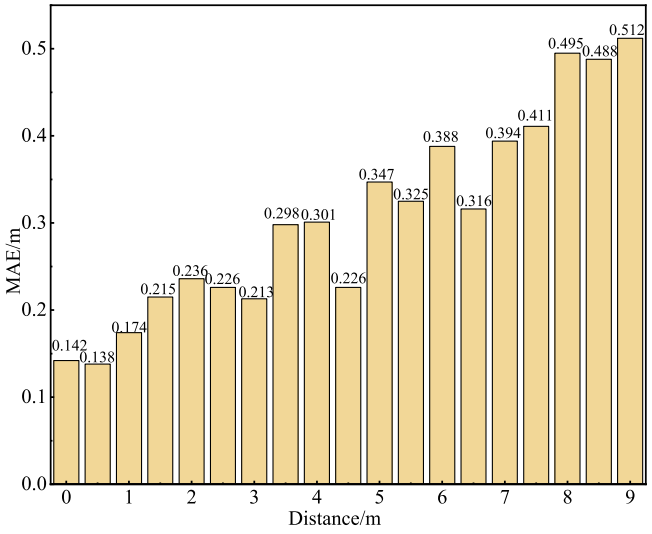


Fig. 13. Predictive effects of ANN.

4.1.4. DBSCAN cluster

DBSCAN is very sensitive to density threshold parameters. According to the actual application requirements, this paper sets the objective function and uses the elbow detection method to automatically obtain the optimal scanning radius under different minimum samples, so as to achieve the optimal optimization of density threshold parameters [31].

To use the elbow detection method, choose the value of z as:

$$z = 2 \times d_E - 1 \tag{11}$$

where d_E indicates the data dimension. And calculate the distance from each point to the z-th closest point, and then sort these distances in ascending order before plotting. Finally find the distance of the inflection point position, which is the value of *Eps*.

The selection method of *MinPts* is

$$MinPts = z + 1 \tag{12}$$

For this paper, the clustering algorithm calculates two-dimensional data, so it can be determined that $z = 3$ and $MinPts = 4$. The value of *Eps* is determined according to the inflection point.

4.2. Positioning results and error analysis

4.2.1. Initial positioning results

The electric arc fault locations in this study are set as shown in Fig. 2, with a total of 10 positions. The locations of the fault points are exactly the same as those in Fig. 2. In Fig. 15, five layout configurations are discussed and analyzed. Firstly, the three antennas with the strongest signals in the redundant antenna array are determined, and then the ellipse algorithm is introduced to obtain the positioning results. In these figures, "×" represents the actual coordinates of the electric arc, while circular markers indicate the predicted coordinates obtained by the algorithm. To simulate the arc after the PV array is connected to the grid, data are collected when the arc is stably burning. It is worth noting that data is collected within a two-second interval, with 10 consecutive five-millisecond data segments obtained. The average value is calculated for each set of data segments, and this process is repeated for 10 experiments under different irradiance levels.

4.2.2. Error analysis

The antennas selected by the algorithm are listed in Table 6. In most experimental scenarios, three antennas that receive the strongest EMR signals are also the closest to the arc. However, due to the prediction errors of ANN, shadow fading, multi-path effects, and site errors, the algorithm selects two sets of antennas in the 4R, 4D, and 5R layouts at position ⑥. Analyze the situation of the 4R layout: in 10 experiments, there are four cases when the antennas S_1, S_2 and S_3 receive the strongest signals, and there are six cases when the antennas S_2, S_3 and S_4 receive the strongest signals. The reason for choosing two sets of antennas in the other two cases is the same. The above situation reflects the flexibility of the algorithm in the selection of antennas. It is worth mentioning that at location ⑥ in the 5D layout, the algorithm shows that it could not be located because the algorithm selects the antennas S_2, S_4 and S_5, but the coordinate connections of this set of antennas form a straight line. The ellipse algorithm based on the trilateration method fails. In such case,

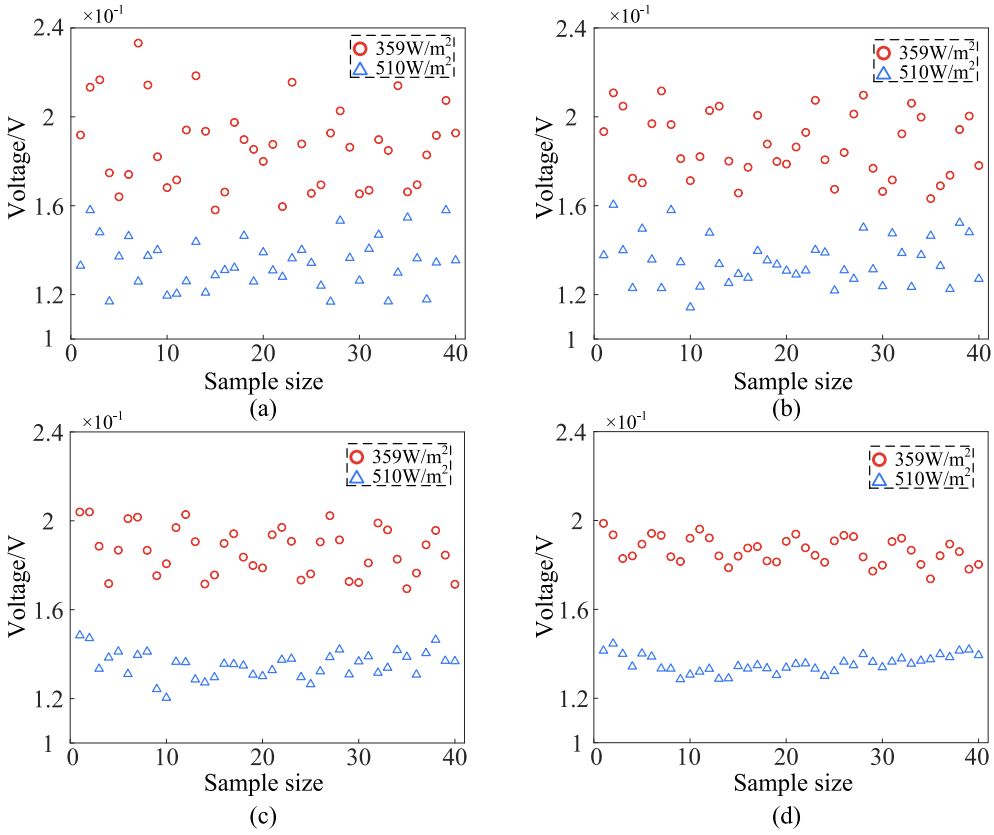


Fig. 14. Distribution of data points in different time windows. (a) 1 ms (b) 2.5 ms (c) 5 ms (d) 7.5 ms.

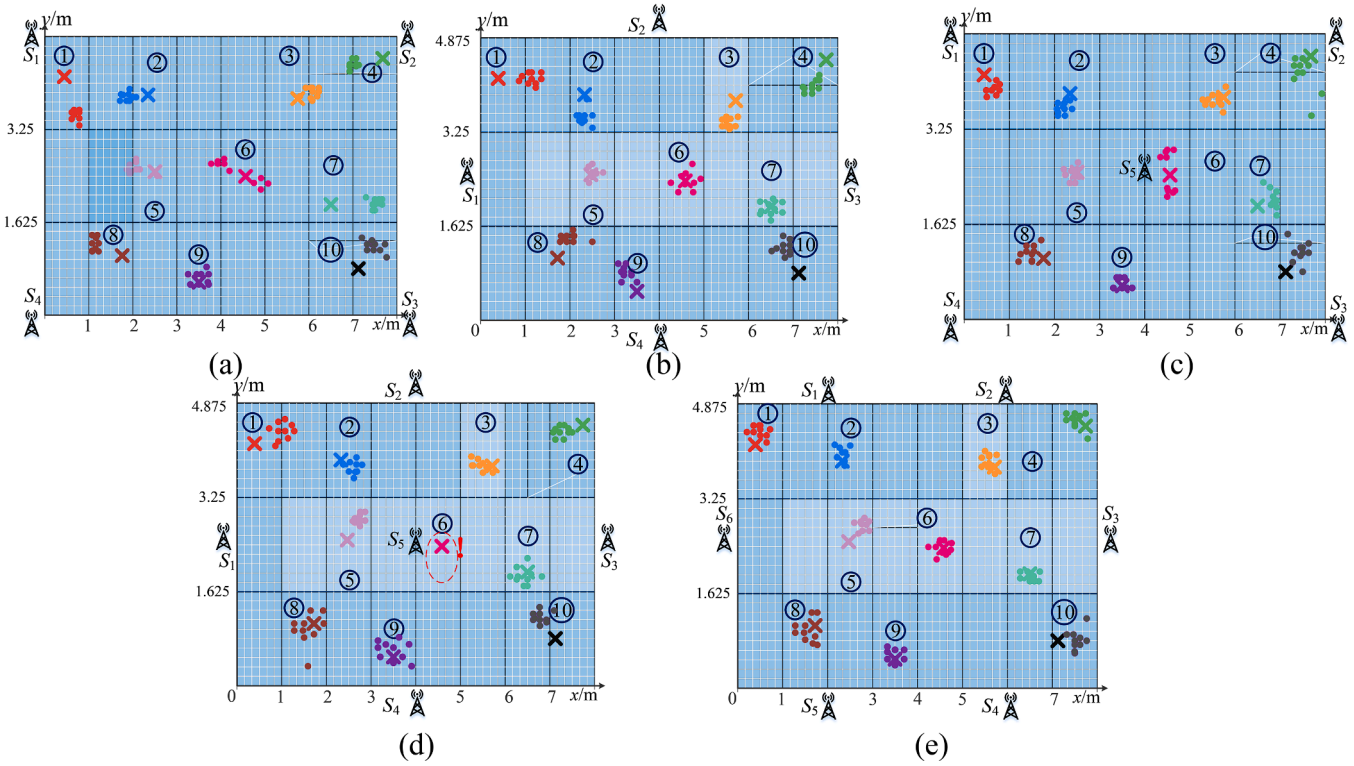


Fig. 15. Preliminary location results of arc fault in PV system. (a) 4R (b) 4D (c) 5R (d) 5D (e) 6H.

Table 6
Antenna numbers for five layouts at 10 locations.

Location	4R	4D	5R	5D	6H
①	S_1,S_2,S_4	S_1,S_2,S_4	S_1,S_4,S_5	S_1,S_2,S_5	S_1,S_5,S_6
②	S_1,S_2,S_4	S_1,S_2,S_4	S_1,S_4,S_5	S_1,S_2,S_5	S_1,S_2,S_6
③	S_1,S_2,S_3	S_2,S_3,S_4	S_1,S_2,S_5	S_2,S_3,S_5	S_1,S_2,S_3
④	S_1,S_2,S_3	S_2,S_3,S_4	S_2,S_3,S_5	S_2,S_3,S_5	S_2,S_3,S_4
⑤	S_1,S_2,S_4	S_1,S_2,S_4	S_1,S_4,S_5	S_1,S_2,S_5	S_1,S_5,S_6
⑥	S_1,S_2,S_3 S_2,S_3,S_4	S_1,S_2,S_4 S_2,S_3,S_4	S_1,S_2,S_5 S_2,S_3,S_5	Unable to locate	S_2,S_4,S_6
⑦	S_2,S_3,S_4	S_2,S_3,S_4	S_2,S_3,S_5	S_3,S_4,S_5	S_2,S_3,S_4
⑧	S_1,S_3,S_4	S_1,S_2,S_4	S_1,S_4,S_5	S_1,S_4,S_5	S_4,S_5,S_6
⑨	S_1,S_3,S_4	S_1,S_2,S_4	S_3,S_4,S_5	S_1,S_4,S_5	S_4,S_5,S_6
⑩	S_2,S_3,S_4	S_2,S_3,S_4	S_2,S_3,S_5	S_3,S_4,S_5	S_2,S_3,S_4

the algorithm discards antenna S_2 (taken in ascending order) and selects the fourth-ranked E_{rms} antenna in descending order, which is S_3.

Ultimately, the algorithm chooses antennas S_3, S_4 and S_5 for localization. For the 10 selected positions in this paper, all antennas have a sufficient utilization rate, indicating that the antenna layout is reasonable.

The MAE of five antenna layouts at 10 locations are shown in Fig. 16, and the real distance between two points is measured by Euclidean distance. In this paper, an antenna is added at the center of the PV array, which effectively reduces the error. Compared with the layouts of 4R and the 5R, there are seven locations where the error of 4R is significantly greater than that of 5R, and the error of 4R is greater than 1 m at location ① and ⑦. The error of 5R at location ①, ④ and ⑩ is greater than 0.5 m, and the error even reaches 0.989 m at location ④. The reason is that in the positioning process, the electromagnetic factors in the surrounding environment are unstable, which leads to deviation in the ANN fitting, resulting in noise points in the positioning results and increases in MAE. This is also one of the reasons for using clustering algorithms: to remove noise points and further improve positioning accuracy. Compared with 4D and 5D, there are five locations where the error of 4D is greater than 5D. The two layouts are both interfered by noise at location ⑧, with errors of 1.148 m and 0.692 m, respectively. The newly added antenna is located at the center of the array, and the probability of being affected by obstacles when receiving EMR signals is smaller than that of other locations, which is also confirmed by the field investigation of obstacles in Fig. 8. Therefore, it can be inferred that increasing the number of antennas at appropriate locations is conductive to improve positioning accuracy.

The location of the fault occurrence also affects the localization accuracy. For the corner locations of the array (①, ④, and ⑩), 6H has the smallest error. Next for the edge locations of the array (②, ③, ⑧ and ⑨), 5R and 6H have the smallest errors, and the variance is also the most stable. Eventually, for the central locations of the array (⑤, ⑥ and ⑦), 5R has the smallest error.

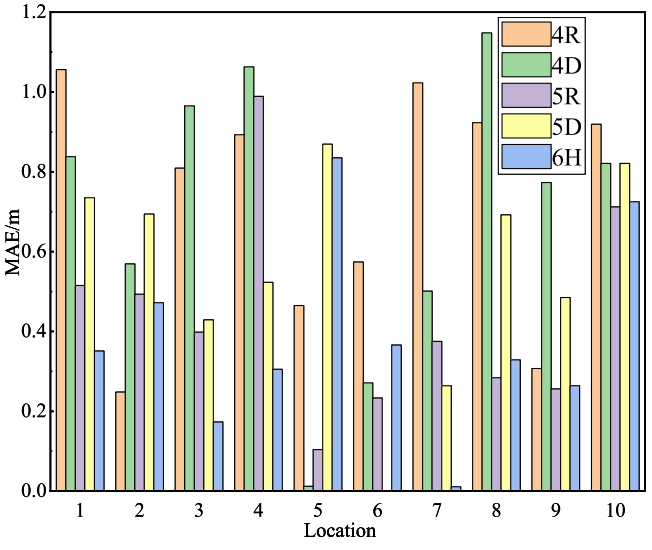


Fig. 16. Comparison of initial positioning error.

To explore the impact of the number of antennas on localization accuracy, this paper conducts 10 experiments at the 10 positions in Fig. 2 using the three-antenna layout (refers to as 3T) and repeats them for six different layout configurations, including 3T. The average localization error, as shown in Table 7, reveals that the MAE decreases as the number of antennas increases. It also illustrates the impact of layout configurations on the error. From both an economic and accuracy perspective, five antennas are sufficient to meet localization requirements.

In summary, the 5R layout of redundant antenna array can minimize positioning error, and the majority of positional error is within 0.5 m.

4.2.3. Cluster results

The EMR signals received by antenna S_1 at location ⑤ in the 5R layout are displayed in Fig. 17, including four possible sampling categories.

In sampling category I, the collected data goes through two stages of arc ignition and stable combustion. In sampling category II, the data only contains information about arc steady combustion. While sampling category III comprises data from both the stable burning and extinguishing. If the arc is extinguished quickly, there is a sampling category IV, which includes the arc generation, steady combustion, and extinction process. For each of these four-sampling category, 10 waveforms are randomly selected, and the average E_{rms} is computed. The same processing is applied to the EMR signal data received by the other four antennas. The proposed algorithm is used for localization in 10 iterations, and the results are depicted in Fig. 18(a).

The algorithm selects antennas S_1, S_4 and S_5 for localization. Both sampling category III and IV include data from the arc extinguishing phase, leading to a lower average E_{rms} than that of sampling category I and II. This results in a reduction in the distance predicted by the ANN. However, the ellipse algorithm exhibits a degree of adaptability to this situation (as seen in Fig. 7(d)), so the localization errors for sampling category III and IV remain within a manageable and reasonable range. Additionally, it can be verified that there is no significant difference between data from the arc ignition phase and the arc stable burning phase. DBSCAN clustering analysis is applied to the four categories in Fig. 17 to obtain the final localization points. The clustering strategy involves identifying the largest cluster and determining its centroid coordinates. The localization errors (Euclidean distance) between the estimated points and the actual fault points are listed in Table 8.

To provide a more detailed demonstration of the clustering results, analysis is conducted on other experiment outcomes as well.

A comparison is made regarding the use of clustering algorithm on the data from sampling category II, as shown in Fig. 17. Fig. 18(b) illustrates the error fluctuations for 10 waveform segments. Among these, the maximum error can reach 0.412 m, while the minimum error can be as low as 0.018 m, indicating that the absolute values of errors vary significantly. However, after applying the clustering algorithm, the errors stabilize at 0.243 m. It is evident that the use of DBSCAN effectively reduces error fluctuations.

Taking the example of location ④ in the 5R layout. In Fig. 15(c), there are two noise points at location ④. This indicates that out of the 10 localization experiments, eight results in accurate localization, while two deviates from the fault point due to noise interference. After

Table 7
Average error for six layouts at 10 locations.

Layout	3T	4R	4D	5R	5D	6H
Average error/m	1.426	0.722	0.696	0.436	0.613	0.383

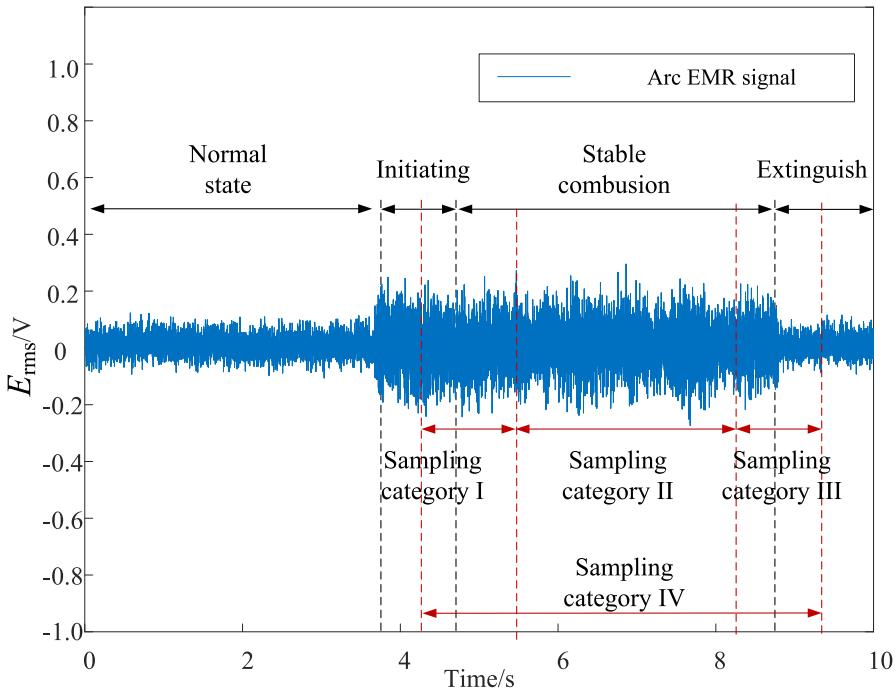


Fig. 17. Analysis of sampling category.

clustering, the results obtained are shown in Fig. 18(c). It can be observed that the DBSCAN clustering algorithm effectively distinguishes core points from noise points. Two data clusters are present in the graph, and clustering results in two cluster centroids. The centroid of the largest cluster is selected as the final localization result, resulting in a localization error of 0.468 m. This represents a reduction of 0.521 m compared to the situation before clustering.

For the special case when the algorithm selects two sets of antennas, this study discusses the scenario at location ⑥ in the 5R layout. In 10 experiments, the algorithm alternately chooses antennas S_2, S_3 and S_5 for five times. In this case, the algorithm could not determine the largest cluster. To address this problem, the algorithm is adjusted as follows: if two clusters contain an equal number of data points, the centroids of both clusters are calculated separately, and then the center coordinates of these two centroids as the final fault location (as shown in Fig. 18(d)). The localization error for Cluster I is 0.346 m, for Cluster II is 0.420 m, and the final localization error is 0.233 m. The clustering results can also be visually observed in the graph. Therefore, the adjusted algorithm yields improve localization performance.

For the location ⑥ in the 4R layout, the algorithm similarly selects two sets of antennas. According to the rule of choosing the largest cluster, in this case, the data from the set of antennas S_1, S_2 and S_3 are discarded, and the data from the set of antennas S_2, S_3 and S_4 are used instead. If the algorithm were to follow the adjusted approach of using both clusters' data to calculate, it would indeed reduce the error by moving the estimated location closer to the actual fault point. However, this would undoubtedly increase the algorithm's complexity, and there wouldn't be a consistent standard for determining when to use data from both clusters. Therefore, the algorithm continues to follow the previous rule.

4.3. Comparative analysis of algorithm

4.3.1. Algorithm validity verification

In order to verify the effectiveness of the proposed algorithm, this paper conducts a comparative analysis of three location algorithms: (I) 3T layout (no redundant antenna array is adopted), using only ellipse algorithm to calculate the arc coordinates. (II) 5R layout (with redundant antenna array), again using ellipse algorithm to calculate the arc coordinates. (III) 5R layout (with a redundant antenna array), using ellipse algorithm for initial localization and then employing clustering analysis to determine the final fault location. In 10 experiments, noise sources similar to the frequency band of EMR signals are added around AFG one or two times to simulate real-world noise interference during the localization process, verifying the feasibility of the clustering algorithm. The MAE for the three methods is shown in Fig. 19.

4.3.2. Algorithm adaptability analysis

In order to validate the adaptability of the proposed algorithm, experiments are conducted using different inverters which are investigated for grid-connected power generation. The parameters of the two inverters are compared in Table 9. Meanwhile, the experiments are repeated at 10 different locations in Fig. 2, with 5R layout. Both inverters utilize the same training dataset, and the results are presented in Table 10.

It can be seen from Table 10 that the error of inverter I is only greater than that of inverter II at location ①, and smaller than that of inverter II at other locations. At location ②, ③, ⑤, ⑧ and ⑨, the difference between the two inverters is relatively small, while the difference between location ④, ⑥, ⑦ and ⑩ is relatively large, with location ④ being the most obvious, possibly ANN has not learned new features. The average error of inverter I is 0.365 m, and that of inverter II is 0.524 m. When different inverters are used to locate arc faults, the error increases without retraining the prediction model, but only slightly larger than 0.5 m. The algorithm can still accurately identify the fault location and demonstrate good adaptability.

4.3.3. Comparison of positioning algorithms

This paper compares the methods in other five references from the perspectives of physical quantities, positioning range, and other aspects, in order to demonstrate the application value and excellent positioning capability of the proposed algorithm. The comparison results are shown in Table 11.

On one hand, after spectral analysis of capacitive current, faults between components, busbars and loads can be located, making it universal in applications [13]. However, it struggles to pinpoint faults

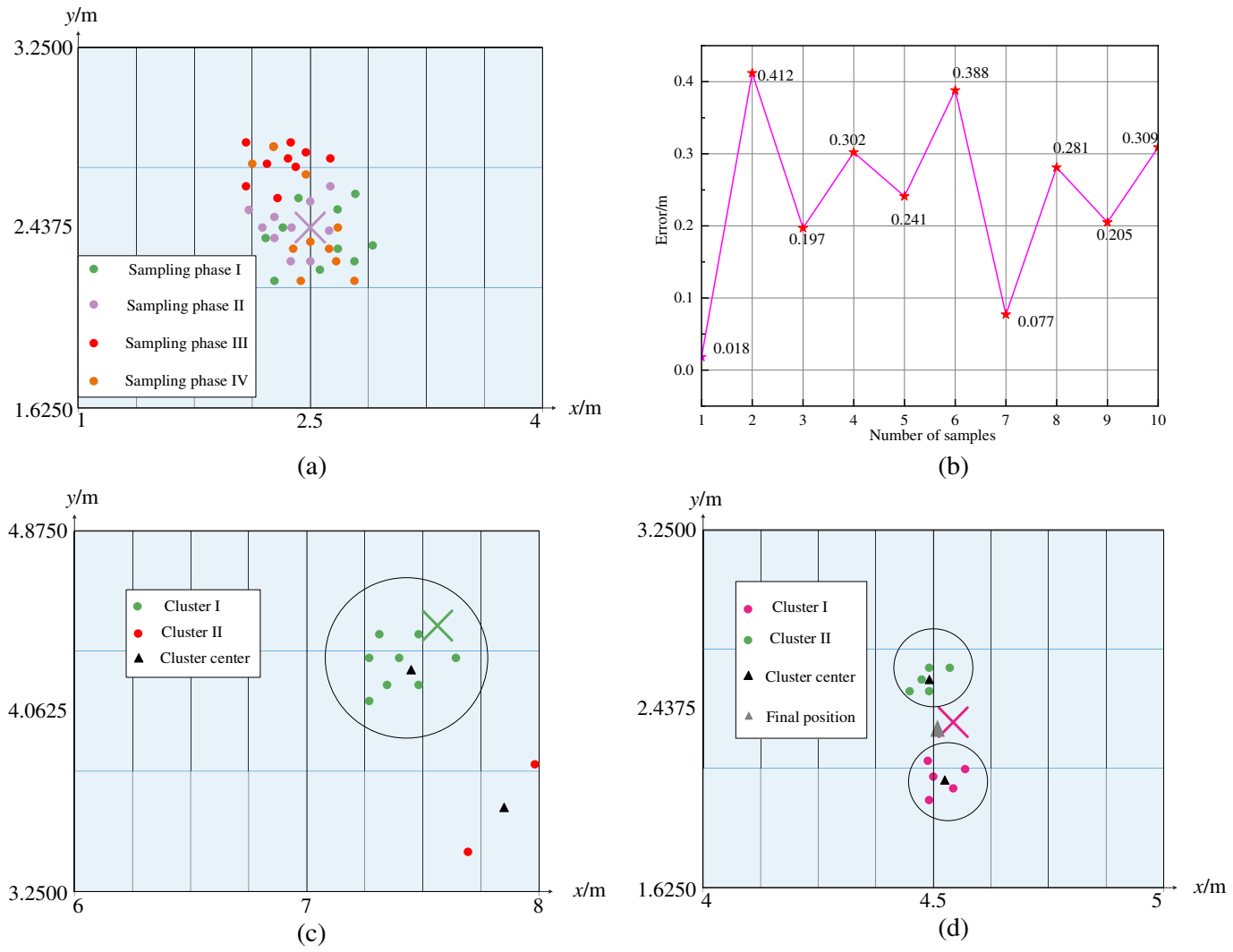


Fig. 18. Positioning results. (a) Comparison of positioning results at different sampling categories (b) Error fluctuations at sampling category II (c) Clusters analysis with noise interferences (d) Cluster analysis results of two sets of antennas.

Table 8
Comparison of positioning results error at different sampling categories.

Sampling category	I	II	III	IV
Error/m	0.269	0.243	0.513	0.305

within the series accurately. On the other hand, the algorithm proposed in [15] offers a positioning range limited to the DC bus, displaying a small average error. This capability effectively mitigates the impact of interference noise in the reflected signal but falls short in accurately locating faults between PV modules. The algorithm's immunity to cope with inverter switching noise characteristics is addressed in [18]. The limited positioning range mentioned in [17,18,22] reflects instances when the experiments are conducted in restricted areas without considering the entire PV array. Furthermore, even in [17], the experiments only utilize PV simulators, neglecting potential scenarios involving arc faults in actual PV arrays. Additionally, the impact of irradiance on positioning accuracy is not considered in [13,17,18].

In contrast, the algorithm introduced in this paper operates at a low sampling rate yet offers a wide-ranging capability to locate most areas within the PV array. This paper takes into account the impact of irradiance on the EMR signal. The algorithm exhibits an average error of 0.365 m, which may seem slightly larger when compared to other

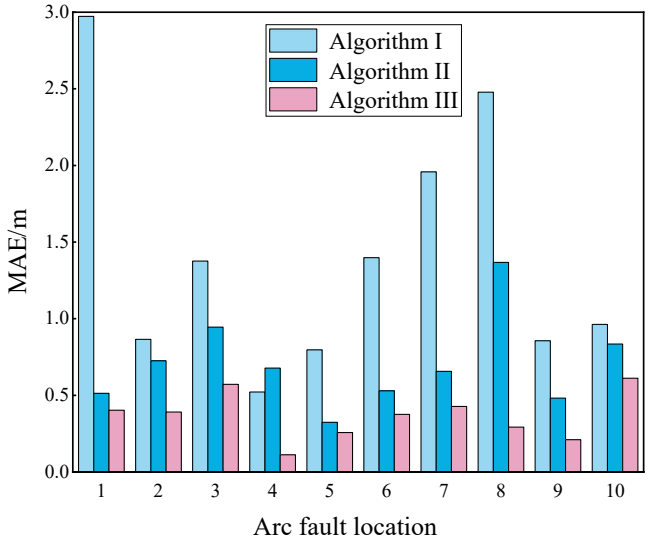


Fig. 19. Results of ablation expression.

Table 9
Comparison of inverter models.

Model	SG6KTL-MT	SGN5400TL
Type	Three-phase	Single-phase
Starting voltage	250 V	90 V
MPPT voltage adjustment range	200 ~ 1000 V	70 ~ 530 V
Switching frequency	16 kHz	19.2 kHz

Table 10
Comparison of positioning errors between different inverters.

Location	Inverter I	Inverter II	Location	Inverter I	Inverter II
①	0.403 *m*	0.394 *m*	⑥	0.376 *m*	0.714 *m*
②	0.391 *m*	0.527 *m*	⑦	0.428 *m*	0.613 *m*
③	0.572 *m*	0.639 *m*	⑧	0.293 *m*	0.413 *m*
④	0.112 *m*	0.538 *m*	⑨	0.210 *m*	0.263 *m*
⑤	0.257 *m*	0.312 *m*	⑩	0.612 *m*	0.827 *m*

references. Notably, in [22], experimental conditions are similar to those in this paper, the error is smaller than that of the proposed algorithm. However, it's essential to consider that the antenna used in this paper possesses a longer receiving range and a more extensive positioning range, resulting in superior actual performance compared to [22]. Additionally, this paper delves into the source of positioning data, highlighting the algorithm's resilience to accidental EMR interference signals and unstable arcing state data. Furthermore, this paper includes an analysis of antenna array layout and quantity, which offers significant advantage over other algorithms.

4.3.4. *Application of the proposed method in other practical PV power stations*

Positioning accuracy exhibits directionality, with higher accuracy achieved when the discharge source is located near the normal line of the antenna pair with the maximum spacing. This has also been corroborated by the error analysis presented in previous chapters. Taking into account various factors such as experimental equipment, the 5R layout emerges as the optimal solution for achieving the required positioning accuracy in the PV field studied in this research. For smaller-scale PV fields where the impact of fading is relatively minor, four antennas may suffice (to demonstrate redundancy, more than three antennas are used).

For larger-scale PV fields, a larger antenna array can be constructed using the 5R layout as the basic configuration. The server parameters used in this paper are: CPU I7-12700H, GPU 1 RTX3050, RAM 24G. The time required for a single fault localization is approximately 1.152 s, indicating that the computational requirements are not demanding and can be handled by an embedded microcontroller. In practical applications, a distributed sampling terminal can be designed, consisting of modules such as antennas, AD (analog to digital) sampling modules, and MCU (microcontroller unit) module. Each measurement point is equipped with a distributed sampling terminal, which is connected to an edge computing terminal via a bus system. The edge computing terminal runs the proposed algorithm to predict the fault location, as illustrated in Fig. 20.

When an arc fault occurs, a subset of the distributed sampling terminals captures the electromagnetic signal of the arc and feeds it back to the edge computing terminal. The edge computing terminal selects data from the five antennas with the strongest signals to determine the fault block. Subsequently, the proposed method is utilized to pinpoint the exact location of the fault.

4.3.5. *Influence of environment temperature on EMR signals*

This section analyzes the impact of environment temperature and objectively evaluates it through detailed experimental data. The relevant experiment settings are as follows: the experiment was conducted from March 21st to March 24th, 2024, during the spring season when there were significant temperature fluctuations within a day. An environment thermometer was applied as the measurement tool. And the

Table 11
Comparison of localization algorithms.

References	Physical quality	Location range	Consider irradiance?	Average error/m	Anti-interference?	Consider antenna layout?
Proposed method	EMR	Array	Yes	0.365	Yes	Yes
[13]	Capacity current	No limit	No	–	Yes	No
[15]	TD	Bus	Yes	0.344	Yes	No
[17]	EMR	Limited	No	0.5	Yes	No
[18]	EMR	Limited	No	0.352	No	No
[22]	EMR	Limited	Yes	0.306	No	No

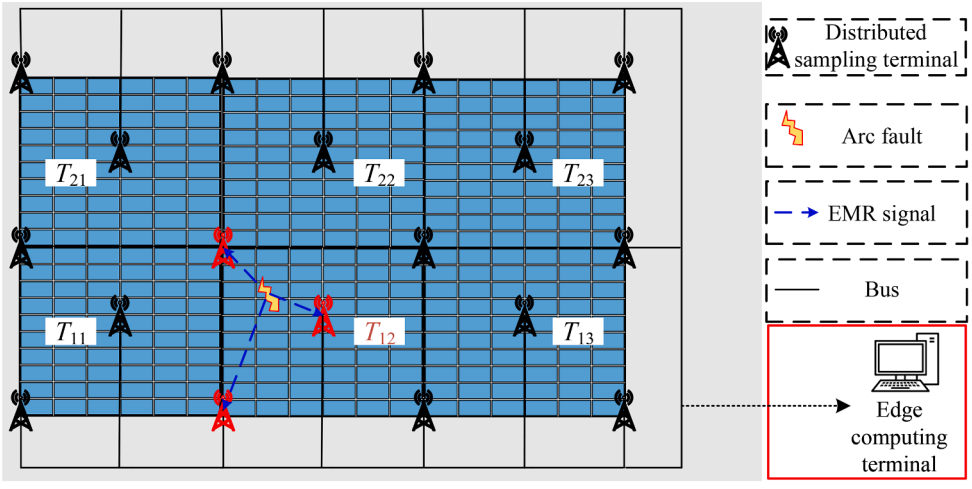


Fig. 20. Schematic diagram of each monitoring module layout in PV power station.

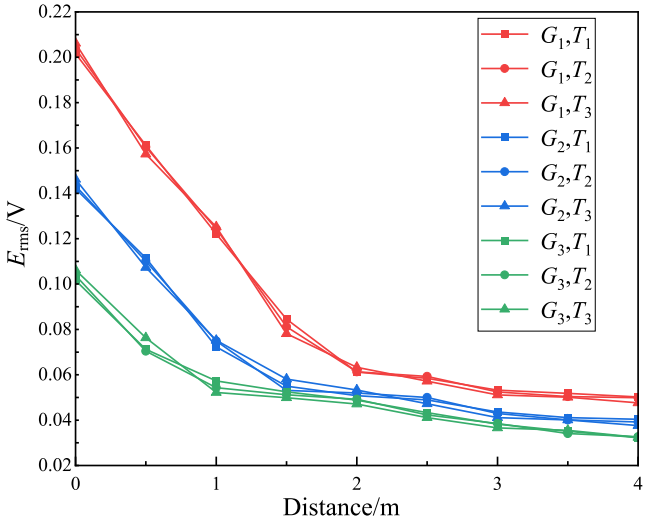


Fig. 21. Influence of temperature on arc EMR signals under different irradiance levels and distances.

experimental site was chosen as the open area described in Section 3.2.1. In addition, signal measurement distances were set from 0 to 4 m, with measurements taken every 0.5 m.

In Fig. 21, from top to button, three clusters of curves are depicted, representing different irradiance ranges in different colors, named G_1, G_2 and G_3 respectively. These clusters of curves are further distinguished by different types of symbols for various temperature ranges, named T_1, T_2 and T_3. Specifically, G_1 represents the irradiance range of 300 ~ 320 W/m^2, G_2 represents the irradiance range of 592 ~ 606 W/m^2, and G_3 represents the irradiance range of 937 ~ 949 W/m^2. Similarly, T_1 represents the temperature range of 9 ~ 11 ℃, T_2 represents the temperature range of 16 ~ 18 ℃, and T_3 represents the temperature range of 26 ~ 28 ℃.

It can be observed that as the distance varies from 0 to 4 m, the decreasing trend of E_{rms} aligns with the description of the LNSM model. The closer the distance between the arc and the antenna, the higher the EMR intensity. Especially within the range of 0 to 2 m, there is a significantly difference in the attenuation rate of EMR signals, with G_1 exhibiting the fastest attenuation, followed by G_2, and then G_3. However, within the range of 2 to 4 m, the rate of signal attenuation is relatively gradual. Additionally, under the same distance conditions, the influence of irradiance on signal strength is significantly greater than that of temperature. This implies that irradiance is a more critical factor in the propagation of EMR signals. When the distance and irradiance are the same, that is, with other factors held constant and only considering a single variable, the impact of temperature variation on EMR signal propagation is relatively smaller.

For further analysis, the standard deviation of E_{rms} under three temperature conditions at each distance is calculated, as shown in Table 12. It can be observed that as the temperature increases, signal fluctuations intensify. However, this fluctuation primarily affects the stability and reliability of the signal rather than its average strength. In fact, examining the *I-V* curve of PV cells, it is evident that irradiance primarily influences the short-circuit current, while temperature mainly affects the open-circuit voltage, with a negligible impact on output current [32]. Additionally, the attenuation formula for EMR signals due to atmospheric temperature is typically measured in larger distance units such as kilometers [33]. This suggests that attenuation becomes a relatively significant factor only when the total area of a PV power plant reaches a certain scale. For the smaller distance range involved in this experiment, such attenuation can be neglected.

Table 12
E_{rms} (V) standard deviation at different temperatures.

Distance/m	T_1	T_2	T_3
0	0.1267	0.2314	0.3382
0.5	0.1352	0.2441	0.3473
1	0.1483	0.2499	0.3477
1.5	0.1637	0.2552	0.3504
2	0.1691	0.2726	0.3772
2.5	0.1962	0.2803	0.3859
3	0.2177	0.2869	0.3884
3.5	0.2401	0.3001	0.4018
4	0.2538	0.3018	0.4155

In summary, although environment temperature has a certain impact on the propagation of EMR signals (primarily manifesting in signal fluctuations), its significance is far less than that of irradiance in influencing the strength of EMR signals. In practical applications, it is crucial to pay more attention to changes in irradiance to enhance the strength and stability of EMR signals.

4.3.6. *Influence of current level through AFG on fault localization results*

The UL1699B standard incorporates arc tests that involve seven controllable parameters, including current, voltage, electrode gap, arcing rate, etc. [34]. However, this standard employs controllable DC sources rather than PV cells. In a fixed-scale PV array, parameters such as maximum power point voltage, current, and impedance are not freely controllable. Therefore, considering the experimental conditions, controlled experiments that align with the UL1699B standard as closely as possible to analyze the impact of arcs generated by different levels of current on localization results should be conducted. During the experiments, the primary parameters under control are I_{mppt} (arc current under MPPT conditions), *Rate* (arcing rate of the AFG) and *Gap* (electrode gap of the AFG).

The experimental design is outlined as follows: The first to third groups of experiments utilized a single-string PV module. The fourth group also employed a single-string module but under conditions of extremely low irradiance to investigate scenarios where the arc may not burn stably. The fifth group, on the other hand, employed a double-string PV module. The columns labeled Head, Middle, and End represent the MAE of the localization results. For the single-string experiments, the beginning, middle, and end of the string correspond to positions ④, ② and ⑤ in Fig. 2, respectively. For the double-string experiment, they correspond to positions ④, ⑧ and ⑥ in Fig. 2. Other experimental settings are identical to those described in Section 4.4.1, and the localization results are summarized in Table 13.

Comparing the first three groups of experiments, it is evident that despite differences in current levels (or irradiance) and arcing rates in the AFG between the first and second groups, as well as variations in the gap in the third group, the localization results did not exhibit significant errors. This indicates that the localization algorithm maintains relative stability despite changes in irradiance, arcing speed, and arc gap. In the fifth group of experiments, the use of higher irradiance levels and a parallel connection of two strings, coupled with a higher ambient temperature, resulted in smaller EMR signals with greater fluctuations compared to the previous experimental results. As for the fourth group of experiments, the lower irradiance levels led to significantly reduced current levels, resulting in intermittent arc combustion. In such scenarios, the EMR signals were highly unstable, leading to larger localization errors, which is a limitation of the proposed method. In other words, when the combustion is stable, the current level has little effect on the localization result.

5. Conclusion

This study introduces a new method for locating DC arc faults for PV array. After a lot of experiments, the following conclusions are drawn:

The ellipse algorithm can effectively handle cases with no solution or multiple solutions in the trilateration method. By employing the

Table 13
Localization results at different positions under different current levels.

Group	I_{mppt}/A	*Rate*/(mm·s^{-1})	*Gap*/mm	*G*/[W·m^{-2}]	*T*/℃	Head/m	Middle/m	End/m
1	3.2	2.4	1.0	476	18	0.935	0.468	0.119
2	7.9	5.1	1.0	839	20	0.926	0.470	0.125
3	8.3	4.9	2.5	811	21	0.944	0.451	0.127
4	2.1	5.0	2.4	248	14	1.331	0.912	1.260
5	15.4	5.1	1.0	1038	27	0.933	0.287	0.228

redundant antenna array, an optimal solution for the number and layout of antennas is derived, known as the 5R layout. DBSCAN has the advantages of eliminating interference from accidental noise and unstable arc data.

Ablation experiment showcases the progressive increase in positioning accuracy, achieving an average positioning error as low as 0.365 m. The proposed method shows novelty, adaptability, and stability.

However, it's important to note that the algorithm's effectiveness and adaptability should still be validated in other types of PV installations, such as large-scale ground PV power stations, industrial and commercial distributed PV power stations, and household PV power stations. In reality, the accuracy of the ANN model needs further improvement, and more comprehensive consideration of environmental factors should be required. Therefore, the next steps involve quantitatively analyzing the optimal antenna layout, antenna quality, and antenna selection for various PV field specifications from a theoretical standpoint. The goal is to explore the key factors that influence the signal of DC arc EMR in PV systems and solve them.

CRediT authorship contribution statement

Liangshi Lin: Investigation, Methodology, Software, Validation, Writing – original draft. **Wei Gao:** Conceptualization, Writing – review & editing. **Gengjie Yang:** Project administration, Supervision.

Acknowledgements

This work is financial supported by the Natural Science Foundation of Fujian, China under Award 2021 J01633.

References

[1] Z. He, P. Chu, C. Li, K. Zhang, H. Wei, Y. Hu, Compound fault diagnosis for photovoltaic arrays based on multi-label learning considering multiple faults coupling, Energy Convers. Manag. 279 (2023) 116742.
[2] J.V. Gompel, D. Spina, C. Develder, Satellite based fault diagnosis of photovoltaic systems using recurrent neural networks, Appl. Energy 305 (2022) 117874.
[3] A. Mellit, An embedded solution for fault detection and diagnosis of photovoltaic modules using thermographic images and deep convolutional neural networks, Eng. Appl. Artif. Intell. 116 (2022) 105459.
[4] X. Chen, K. Ding, H. Yang, X. Chen, J. Zhang, M. Jiang, R. Gao, Z. Liu, Research on real-time identification method of model parameters for the photovoltaic array, Appl. Energy 342 (2023) 121157.
[5] K. Jia, C. Gu, L. Li, Z. Xuan, T. Bi, D. Thomas, Sparse voltage amplitude measurement based fault location in large-scale photovoltaic power plants, Appl. Energy 211 (2018) 568–581.
[6] X. Chen, W. Gao, C. Hong, Y. Tu, A novel series arc fault detection method for photovoltaic system based on multi-input neural network, Int. J. Electr. Power Energy Syst. 140 (2022) 108018.
[7] S. Lu, B.T. Phung, D. Zhang, A comprehensive review on DC arc faults and their diagnosis methods in photovoltaic systems, Renew. Sust. Energ. Rev. 89 (2018) 88–98.
[8] W. Gao, H. Jin, G. Yang, Series arc fault diagnosis method of photovoltaic arrays based on GASF and improved DCGAN, Adv. Eng. Inform. 54 (2022) 101809.
[9] S. Lu, R. Ma, T. Sirojan, B.T. Phung, D. Zhang, Lightweight transfer nets and adversarial data augmentation for photovoltaic series arc fault detection with limited fault data, Int. J. Electr. Power Energy Syst. 130 (2021) 107035.
[10] H. Qian, B. Lee, Z. Wu, G. Wang, Research on DC arc fault detection in PV systems based on adjacent multi-segment spectral similarity and adaptive threshold model, Sol. Energy 264 (2023) 112011.
[11] W. Wang, B. Xu, G. Zou, D. Liang, Arc fault detection method based on voltage characteristic energy amplitude and phase mapping distribution distances, Electr. Power Syst. Res. 225 (2023) 109866.
[12] D.W. Upton, K.K. Mistry, P.J. Mather, Z.D. Zaharis, R.C. Atkinson, C. Tachtatzis, P. I. Lazaridis, A review of techniques for RSS-based radiometric partial discharge localization, Sensors 21 (3) (2021) 909.
[13] Q. Xiong, S. Ji, X. Liu, X. Feng, F. Zhang, L. Zhu, A.L. Gattozzi, R.E. Hebner, Detecting and localizing series arc fault in photovoltaic systems based on time and frequency characteristics of capacitor current, Sol. Energy 170 (2018) 788–799.
[14] D.R. Akbar, E. Prasetyono, D.O. Anggriawan, Identification of series DC arc fault location in MPPT photovoltaic system based on fast Fourier transform and artificial neural network, in: 2022 International Electronics Symposium (IES), 2022, pp. 26–33.
[15] C. Wu, Y. Hu, Z. Li, F. Wang, On-line detection and location of DC bus arc fault in PV systems based on SSTDR, Proceedings of the CSEE 40 (8) (2020) 2725–2735.
[16] X. Li, H. Wang, P. Guo, W. Xiong, J. Huang, Series DC arc fault detection and location in wind-solar-storage hybrid system based on variational mode decomposition, Electr. Power Syst. Res. 209 (2022) 107991.
[17] K. Li, S. Zhao, Y. Wang, A planar location method for DC arc faults using dual radiation detection points and DANN, IEEE Trans. Instrum. Meas. 69 (8) (2020) 5478–5487.
[18] S. Zhao, G. You, X. Hou, X. Xu, Y. Zhang, H. An, A spatial location method for DC series arc faults based on RSSI and Bayesian regularization neural network, IEEE Sens. J. 21 (24) (2021) 27868–27877.
[19] M.A. Jangjoo, M. Allahbakhshi, H.R. Mirzaei, UHF sensors positioning on the power transformer tank to enhance the partial discharge localization accuracy, Electr. Power Syst. Res. 218 (2023) 109174.
[20] A. Chaisang, T. Tiengthong, M.M. Maw, S. Promwong, Partial discharge localization model in power transformer with fingerprinting technique, in: 2023 9th International Conference on Engineering, Applied Sciences, and Technology (ICEAST), 2023, pp. 51–54.
[21] W. Gao, W. He, R.J. Wai, X. Zeng, M. Guo, High-impedance arc fault modeling for distribution networks based on dynamic geometry dimension, Electr. Power Syst. Res. 229 (2024) 110109.
[22] H. Jin, W. Gao, L. Lin, G. Yang, Arc fault location method for photovoltaic array based on grid fingerprint matching, High Volt. Engin. 50 (02) (2024) 805–815.
[23] I. Cappelli, F. Carli, A. Fort, M. Intravaia, F. Micheletti, G. Peruzzi, V. Vignoli, Enhanced visible light localization based on machine learning and optimized fingerprinting in wireless sensor networks, IEEE Trans. Instrum. Meas. 72 (2023) 1–10.
[24] Q. Yu, L. Luo, T. Jia, H. Song, G. Sheng, X. Jiang, Spatial partial discharge localization technology in substation based on RSSI statistical analysis of UHF signals, High Volt. Engin. 46 (12) (2020) 4163–4171.
[25] L. Wu, D. He, B. Ai, J. Wang, H. Qi, K. Guan, Z. Zhong, Artificial neural network based path loss prediction for wireless communication network, IEEE Access 8 (2020) 199523–199538.
[26] H.S. Jo, C. Park, E. Lee, H.K. Choi, J. Park. Path loss prediction based on machine learning techniques: principal component analysis, artificial neural network, and gaussian process, Sensors 20 (7) (2020) 1927.
[27] B. Yang, L. Guo, R. Guo, M. Zhao, T. Zhao, A novel trilateration algorithm for RSSI-based indoor localization, IEEE Sens. J. 20 (14) (2020) 8164–8172.
[28] T. Wang, B. Yang, RSSI indoor mobile location based on set membership filtering with event triggering mechanism, Navigation Positioning and Timing 8 (02) (2021) 88–96.
[29] A.A. Kurzhanskiy, P. Varaiya, Ellipsoidal Toolbox (ET), in: Proceedings of the 45th IEEE Conference on Decision and Control, 2006, pp. 1498–1503.
[30] G. Zhang, M. Zhu, Y. Wang, Q. Liu, Y. Li, J. Deng, H. Mu, Research on detection and location of substation area discharge source based on mobile UHF antenna array, Proceedings of the CSEE 37 (10) (2017) 2761–2773.
[31] S. Louhichi, M. Gzara, H.B. Abdallah, A density based algorithm for discovering clusters with varied density, in: 2014 World Congress on Computer Applications and Information Systems (WCCAIS), 2014, pp. 1–6.
[32] M. Zaimi, H.E. Achouby, A. Ibral, E.M. Assaid, Determining combined effects of solar radiation and panel junction temperature on all model-parameters to forecast peak power and photovoltaic yield of solar panel under non-standard conditions, Sol. Energy 191 (2017) 341–359.
[33] I. Expósito, M.G. Sánchez, I. Cuiñas, Computing the influence of environment conditions in electromagnetic measurements uncertainty, IEEE T. Antenn. Propag. 67 (6) (2019) 4084–4090.
[34] K. Park, J. Bang, S. Bang, J. Park, S. Hong, Statistical analysis of ignition time and energy of the TFR-CV cable ignition experiments using 100 W DC arc, J. Electr. Eng. Technol. 17 (2022) 3593–3599.